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DISHWASHER APPLIANCE INCLUDING ALTERNATIVE INPUT MEANS

Abstract

A dishwasher appliance includes a tub; a door movably coupled to the tub between a latched position and an unlatched position; an input module provided on the door, the input module configured to receive one or more input patterns; and a controller operably coupled with the input module, the controller configured to perform an operation. The operation includes determining that the door is moved from the unlatched position to the latched position; receiving, via the input module, a first input pattern of the one or more input patterns after determining that the door is moved from the unlatched position to the latched position; retrieving a first set of cycle settings corresponding to the first input pattern in response to receiving the first input pattern; and initiating a washing cycle after retrieving the first set of cycle settings.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present subject matter relates generally to domestic appliances, and more particularly to alternative input methods for dishwasher appliances.

BACKGROUND OF THE INVENTION

[0002] Dishwashing appliances generally include a tub that defines a wash chamber. Rack assemblies can be mounted within the wash chamber of the tub for receipt of articles for washing. Multiple spray assemblies can be positioned within the wash chamber for applying or directing wash liquid (e.g., water, detergent, etc.) towards articles disposed within the rack assemblies in order to clean such articles. After being applied or directed towards the rack assemblies and/or articles therein, the wash liquid generally flows by gravity to or towards a bottom of the wash chamber, such as to a sump positioned at the bottom of the wash chamber. Dishwashing appliances are also typically equipped with one or more pumps, such as a circulation pump or a drain pump, for directing or motivating wash liquid from the sump to, e.g., the spray assemblies or an area outside of the dishwashing appliance.

[0003] Generally, common dishwasher appliances include a user interface including one or more user inputs for adjusting cycle settings, setting times, initiating washing operations, and the like. In particular, one or more input devices (e.g., buttons) may be provided through which users can adjust settings such as water temperature, wash intensity, washing zones, etc. However, drawbacks exist to current appliances with regards to user inputs. For one example, some interfaces are positioned in difficult to reach areas, such as when the door is in a closed position. For another example, selecting certain wash operations or cycles with customized settings can be difficult or entered incorrectly into the interface.

[0004] Accordingly, a dishwasher appliance that obviates one or more of the above-mentioned drawbacks would be desirable. In particular, a dishwasher appliance with one or more alternative input means would be useful.

BRIEF DESCRIPTION OF THE INVENTION

[0005] Aspects and advantages of the invention will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the invention.

[0006] In one exemplary aspect of the present disclosure, a dishwasher appliance is provided. The dishwasher appliance may include a tub defining a receiving chamber; a door movably coupled to the tub to provide selective access to the receiving chamber, the door being movable between a latched position and an unlatched position; an input module provided on the door, the input module configured to receive one or more input patterns; and a controller operably coupled with the input module, the controller configured to perform an operation. The operation may include determining that the door is moved from the unlatched position to the latched position; receiving, via the input module, a first input pattern of the one or more input patterns after determining that the door is moved from the unlatched position to the latched position; retrieving a first set of cycle settings corresponding to the first input pattern in response to receiving the first input pattern; and initiating a washing cycle after retrieving the first set of cycle settings.

[0007] In another exemplary aspect of the present disclosure, a method of operating a dishwasher appliance is provided. The dishwasher appliance may include a door and an input module. The method may include determining that the door is moved from an unlatched position to a latched position; receiving, via the input module, a first input pattern of among a plurality of input patterns after determining that the door is moved from the unlatched position to the latched position; retrieving a first set of cycle settings corresponding to the first input pattern in response to

receiving the first input pattern; and initiating a washing cycle after retrieving the first set of cycle settings.
[0008] These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0009] A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.
[0010] FIG. **1** provides a perspective view of an exemplary embodiment of a dishwashing appliance of the present disclosure with a door in a partially open position.
[0011] FIG. **2** provides a side, cross-sectional view of the exemplary dishwashing appliance of FIG. **1**.
[0012] FIG. **3** provides a flow chart illustrating a method of operating a dishwasher appliance according to exemplary embodiments of the present disclosure.
[0013] Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.
DETAILED DESCRIPTION
[0014] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.
[0015] As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. The terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). In addition, here and throughout the specification and claims, range limitations may be combined and/or interchanged. Such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other. The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.
[0016] Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “generally,” “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 10 percent margin, i.e., including values within

ten percent greater or less than the stated value. In this regard, for example, when used in the context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction, e.g., “generally vertical” includes forming an angle of up to ten degrees in any direction, e.g., clockwise or counterclockwise, with the vertical direction V.

[0017] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” In addition, references to “an embodiment” or “one embodiment” does not necessarily refer to the same embodiment, although it may. Any implementation described herein as “exemplary” or “an embodiment” is not necessarily to be construed as preferred or advantageous over other implementations. Moreover, each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0018] Turning now to the figures, FIGS. **1** and **2** depict an exemplary dishwasher or dishwashing appliance (e.g., dishwashing appliance **100**) that may be configured in accordance with aspects of the present disclosure. Generally, dishwasher **100** defines a vertical direction V, a lateral direction L, and a transverse direction T. Each of the vertical direction V, lateral direction L, and transverse direction T are mutually perpendicular to one another and form an orthogonal direction system.

[0019] Dishwasher **100** includes a cabinet **102** having a tub **104** therein that defines a wash chamber **106**. As shown in FIG. **2**, tub **104** extends between a top **107** and a bottom **108** along the vertical direction V, between a pair of side walls **110** along the lateral direction L, and between a front side **111** and a rear side **112** along the transverse direction T.

[0020] Tub **104** includes a front opening **114** (FIG. **1**). In some embodiments, the dishwasher appliance **100** may also include a door **116** at the front opening **114**. The door **116** may, for example, be hinged at its bottom for movement between a normally closed vertical position, wherein the wash chamber **106** is sealed shut for washing operation, and a horizontal open position for loading and unloading of articles from dishwasher **100**. A door closure mechanism or assembly **118** may be provided to lock and unlock door **116** for accessing and sealing wash chamber **106**. For instance, door closure mechanism **118** may be a latch or latch system. A latch arm, tab, or protrusion may be positioned on one of door **116** or tub **104** while a latch receptacle or cavity may be positioned on the other of door **116** or tub **104**. Accordingly, door **116** may be selectively movable between an unlatched position (FIG. **1**) and a latched position (FIG. **2**).

[0021] Dishwasher **100** may include a sensor **119**. Sensor **119** may be positioned at door closure mechanism **118**. Accordingly, sensor **119** may be configured to monitor a position of door **116** (e.g., whether door **116** is in the latched or unlatched position). Sensor **119** may be operably connected with a controller (e.g., controller **160**, described below). Sensor **119** may send signals to the controller to indicate the position of the door at certain intervals. Additionally or alternatively, sensor **119** may be configured to emit a signal to the controller any time door **116** is moved from the latched position to the unlatched position, or vice versa. Sensor **119** may be any suitable type of sensor, such as an electronic sensor, a mechanical sensor, an electromechanical sensor, an optical sensor, a magnetic sensor, or the like.

[0022] In exemplary embodiments, tub side walls **110** accommodate a plurality of rack assemblies. For instance, guide rails **120** may be mounted to side walls **110** for supporting a lower rack assembly **122**, a middle rack assembly **124**, or an upper rack assembly **126**. In some such embodiments, upper rack assembly **126** is positioned at a top portion of wash chamber **106** above middle rack assembly **124**, which is positioned above lower rack assembly **122** along the vertical direction V.

[0023] Generally, each rack assembly **122**, **124**, **126** may be adapted for movement between an extended loading position (not shown) in which the rack is substantially positioned outside the wash chamber **106**, and a retracted position (shown in FIGS. **1** and **2**) in which the rack is located inside the wash chamber **106**. In some embodiments, movement is facilitated, for instance, by rollers **128** mounted onto rack assemblies **122**, **124**, **126**, respectively. Although guide rails **120** and rollers **128** are illustrated herein as facilitating movement of the respective rack assemblies **122**, **124**, **126**, it should be appreciated that any suitable sliding mechanism or member may be used according to alternative embodiments.

[0024] In optional embodiments, some or all of the rack assemblies **122**, **124**, **126** are fabricated into lattice structures including a plurality of wires or elongated members **130** (for clarity of illustration, not all elongated members making up rack assemblies **122**, **124**, **126** are shown in FIG. **2**). In this regard, rack assemblies **122**, **124**, **126** are generally configured for supporting articles within wash chamber **106** while allowing a flow of wash liquid to reach and impinge on those articles (e.g., during a cleaning or rinsing cycle). According to additional or alternative embodiments, a silverware basket (not shown) is removably attached to a rack assembly (e.g., lower rack assembly **122**), for placement of silverware, utensils, and the like, that are otherwise too small to be accommodated by the rack assembly.

[0025] Generally, dishwasher **100** includes one or more spray assemblies for urging a flow of fluid (e.g., wash liquid) onto the articles placed within wash chamber **106**. In exemplary embodiments, dishwasher **100** includes a lower spray arm assembly **134** disposed in a lower region **136** of wash chamber **106** and above a sump **138** so as to rotate in relatively close proximity to lower rack assembly **122**.

[0026] In additional or alternative embodiments, a mid-level spray arm assembly **140** is located in an upper region of wash chamber **106** (e.g., below and in close proximity to middle rack assembly **124**). In this regard, mid-level spray arm assembly **140** may generally be configured for urging a flow of wash liquid up through middle rack assembly **124** and upper rack assembly **126**.

[0027] In further additional or alternative embodiments, an upper spray assembly **142** is located above upper rack assembly **126** along the vertical direction V. In this manner, upper spray assembly **142** may be generally configured for urging or cascading a flow of wash liquid downward over rack assemblies **122**, **124**, and **126**.

[0028] In yet further additional or alternative embodiments, upper rack assembly **126** may further define an integral spray manifold **144**. As illustrated, integral spray manifold **144** may be directed upward, and thus generally configured for urging a flow of wash liquid substantially upward along the vertical direction V through upper rack assembly **126**.

[0029] In still further additional or alternative embodiments, a filter clean spray assembly **145** is disposed in a lower region **136** of wash chamber **106** (e.g., below lower spray arm assembly **134**) and above a sump **138** so as to rotate in relatively close proximity to a filter assembly **210** (e.g., FIG. **4**). For instance, filter clean spray assembly **145** may be directed downward to urge a flow of wash liquid across a portion of filter assembly **210** (FIG. **4**) or sump **138**.

[0030] The various spray assemblies and manifolds described herein may be part of a fluid distribution system or fluid circulation assembly **150** for circulating wash liquid in tub **104**. In certain embodiments, fluid circulation assembly **150** includes a circulation pump **152** for circulating wash liquid in tub **104**. Circulation pump **152** may be located within sump **138** or within a machinery compartment located below sump **138** of tub **104**.

[0031] When assembled, circulation pump **152** may be in fluid communication with an external water supply line (not shown) and sump **138**. A water inlet valve **153** may be positioned between the external water supply line and circulation pump **152** (e.g., to selectively allow water to flow from the external water supply line to circulation pump **152**). Additionally or alternatively, water inlet valve **153** may be positioned between the external water supply line and sump **138** (e.g., to selectively allow water to flow from the external water supply line to sump **138**). During use, water inlet valve **153** may be selectively controlled to open to allow the flow of water into dishwasher

100 and may be selectively controlled to close and thereby cease the flow of water into dishwasher **100**. Further, fluid circulation assembly **150** may include one or more fluid conduits or circulation piping for directing wash fluid from circulation pump **152** to the various spray assemblies and manifolds. In exemplary embodiments, such as that shown in FIG. **2**, a primary supply conduit **154** extends from circulation pump **152**, along rear **112** of tub **104** along the vertical direction V to supply wash liquid throughout wash chamber **106**.

[0032] In some embodiments, primary supply conduit **154** is used to supply wash liquid to one or more spray assemblies (e.g., to mid-level spray arm assembly **140** or upper spray assembly **142**). It should be appreciated, however, that according to alternative embodiments, any other suitable plumbing configuration may be used to supply wash liquid throughout the various spray manifolds and assemblies described herein. For instance, according to another exemplary embodiment, primary supply conduit **154** could be used to provide wash liquid to mid-level spray arm assembly **140** and a dedicated secondary supply conduit (not shown) could be utilized to provide wash liquid to upper spray assembly **142**. Other plumbing configurations may be used for providing wash liquid to the various spray devices and manifolds at any location within dishwashing appliance **100**.

[0033] Each spray arm assembly **134** and **140**, upper spray assembly **142**, integral spray manifold **144**, filter clean assembly **145**, or other spray device may include an arrangement of discharge ports or orifices for directing wash liquid received from circulation pump **152** onto dishes or other articles located in wash chamber **106**. The arrangement of the discharge ports, also referred to as jets, apertures, or orifices, may provide a rotational force by virtue of wash liquid flowing through the discharge ports. Alternatively, spray assemblies **134**, **140**, **142**, **145** may be motor-driven, or may operate using any other suitable drive mechanism. Spray manifolds and assemblies may also be stationary. The resultant movement of the spray assemblies **134**, **140**, **142**, **145** and the spray from fixed manifolds provides coverage of dishes and other dishwasher contents with a washing spray. Other configurations of spray assemblies may be used as well. For instance, dishwasher **100** may have additional spray assemblies for cleaning silverware, for scouring casserole dishes, for spraying pots and pans, for cleaning bottles, etc.

[0034] In optional embodiments, circulation pump **152** urges or pumps wash liquid (e.g., from filter assembly **210**) to a diverter **156** (FIG. **2**). In some such embodiments, diverter **156** is positioned within sump **138** of dishwashing appliance **100**). Diverter **156** may include a diverter disk (not shown) disposed within a diverter chamber **158** for selectively distributing the wash liquid to the spray assemblies **134**, **140**, **142**, or other spray manifolds. For instance, the diverter disk may have a plurality of apertures that are configured to align with one or more outlet ports (not shown) at the top of diverter chamber **158**. In this manner, the diverter disk may be selectively rotated to provide wash liquid to the desired spray device.

[0035] In exemplary embodiments, diverter **156** is configured for selectively distributing the flow of wash liquid from circulation pump **152** to various fluid supply conduits—only some of which are illustrated in FIG. **2** for clarity. In certain embodiments, diverter **156** includes four outlet ports (not shown) for supplying wash liquid to a first conduit for rotating lower spray arm assembly **134**, a second conduit for supplying wash liquid to filter clean assembly **145**, a third conduit for spraying an auxiliary rack such as the silverware rack, and a fourth conduit for supply mid-level or upper spray assemblies **140**, **142** (e.g., primary supply conduit **154**).

[0036] Drainage of soiled wash liquid within sump **138** may occur, for instance, through a drain assembly **166** (e.g., during or as part of a drain cycle). In particular, wash liquid may exit sump **138** through a drain outlet and may flow through a drain conduit. In some embodiments, a drain pump **168** downstream of sump **138** facilitates drainage of the soiled wash liquid by urging or pumping the wash liquid to a drain line external to dishwasher **100**. Drain pump **168** may be downstream of a filter (e.g., positioned at or near sump **138**). Additionally or alternatively, an unfiltered flow path may be defined through sump **138** to drain conduit such that an unfiltered fluid flow may pass through sump **138** to drain conduit without first passing through filtration media.

[0037] In certain embodiments, dishwasher **100** includes a controller **160** configured to regulate operation of dishwasher **100** (e.g., initiate one or more wash operations). Controller **160** may include one or more memory devices and one or more microprocessors, such as general or special purpose microprocessors operable to execute programming instructions or micro-control code associated with a wash operation that may include a wash cycle, rinse cycle, or drain cycle. The memory may represent random access memory such as DRAM, or read only memory such as ROM or FLASH. In some embodiments, the processor executes programming instructions stored in memory. The memory may be a separate component from the processor or may be included onboard within the processor. Alternatively, controller **160** may be constructed without using a microprocessor, e.g., using a combination of discrete analog or digital logic circuitry—such as switches, amplifiers, integrators, comparators, flip-flops, AND gates, and the like—to perform control functionality instead of relying upon software. It should be noted that controllers as disclosed herein are capable of and may be operable to perform any methods and associated method steps as disclosed herein.
[0038] Controller **160** may be positioned in a variety of locations throughout dishwasher **100**. In optional embodiments, controller **160** is located within a control panel area **162** of door **116** (e.g., as shown in FIGS. **1** and **2**). Input/output ("I/O") signals may be routed between the control system and various operational components of dishwasher **100** along wiring harnesses that may be routed through the bottom of door **116**. Typically, the controller **160** includes a user interface panel/controls **164** through which a user may select various operational features and modes and monitor progress of dishwasher **100**. In some embodiments, user interface **164** includes a general purpose I/O ("GPIO") device or functional block. In additional or alternative embodiments, user interface **164** includes input components, such as one or more of a variety of electrical, mechanical or electro-mechanical input devices including rotary dials, push buttons, and touch pads. In further additional or alternative embodiments, user interface **164** includes a display component, such as a digital or analog display device designed to provide operational feedback to a user. When assembled, user interface **164** may be in operative communication with the controller **160** via one or more signal lines or shared communication busses. For at least one example, user interface **164** includes an indicator **165**. Indicator **165** may be a clean indicator to indicate that the load within tub **104** is in a clean state (e.g., after a completion of a washing cycle or operation). Indicator **165** may be a single light (e.g., light emitting diode or LED), a pattern of lights, a word, or the like.
[0039] User interface **164** may include an audio module **167**. Audio module **167** may be positioned within door **116**. Additionally or alternatively, audio module **167** may be positioned within cabinet **102**. Audio module **167** may be operably connected with controller **160**. Audio module **167** may be configured to produce, emit, or otherwise play a plurality of tones. Accordingly, audio module **167** may include one or more speakers. The plurality of tones may include, for instance, alerts, notifications, timer buzzers, or other sounds designed to provide users with notifications pertaining to appliance **100**.
[0040] Controller **160** may include a timer. For instance, one or more of the circuitry provided within controller **160** may be configured to perform a time keeping operation. For at least one example, as will be explained further below, controller **160** (via the timer) may keep track of a length of time for which door **116** is in the unlatched or latched position. Additionally or alternatively, the timer may be provided as a separate instrument within dishwasher **100**. Further still, the timer may be incorporated as part of a remote terminal (e.g., a consumer device, described below).
[0041] Additionally or alternatively, appliance **100** may include an input module **200**. Input module **200** may be positioned apart from user interface **164**. For at least one example, input module **200** is provided at or within a handle **170** of door **116**. As shown particularly in FIG. **1**, handle **170** may be provided on an outer front panel of door **116**. However, additional or alternative handles may be incorporated into specific embodiments, such as recessed handles, top handles, dual handles, or the

like. Accordingly, handle **170** is provided by way of example only and the disclosure is not limited to handle **170** shown.

[0042] Input module **200** may be positioned on an outer surface of handle **170**. For instance, input module **200** may be positioned on a front surface of handle **170**. Two or more input modules **200** may be provided on handle **170**, according to some embodiments. For example, a first input module **200** may be positioned at a first lateral side of handle **170** and a second input module **200** may be positioned at a second lateral side of handle **170**. Additionally or alternatively, a decorative logo on or in handle **170** may be or include input module **200**.

[0043] As described, input module **200** may be a touch input. For instance, input module **200** may define one or more discrete capacitive touch zones for input (e.g., on input module **200**). Additionally or alternatively, the touch zone(s) may be provided as input selectors. During operation of appliance **100**, a user's touch or engagement with the discrete touch zones of input module **200** may select or initiate one or more functions of the appliance **100**, such as the mode of operation, water temperature selected, and other relevant information. At input module **200**, variations in capacitance caused by a user's engagement may be detected. In response, user interface **164** may cause one or more functions to be selected or initiated (e.g., via controller **160**-FIG. **1**). Optionally, one or more of the touch zones may be labeled (e.g., via a printed text or design) to indicate one or more functions to which the touch zone corresponds.

[0044] In additional or alternative embodiments, input module **200** may include a microphone **202**. Microphone **202** may include one or more microphone receivers attached thereto. Microphone **202** may be used for monitoring the sound waves, noises, or other vibrations generated by a user near appliance **100**. For example, microphone **202** may be one or more microphones, acoustic detection devices, vibration sensors, or any other suitable acoustic transducers that are positioned at one or more locations in or around appliance **100** (e.g., such as within handle **170**, user interface **164**, etc.). For example, according to exemplary embodiments, microphone **202** may detect any sounds within audible range of appliance **100**. Additionally or alternatively, microphone **202** may be positioned elsewhere within appliance **100**. In this regard, any suitable microphone **202** that is acoustically or electrically coupled with appliance **100** (e.g., controller **160**) may be used to monitor sounds. Microphone **202** may be communicatively coupled to (i.e., in operative communication with) controller **160**.

[0045] In still additional or alternative embodiments, input module **200** may include one or more visual detectors (e.g., such as a camera). For instance, input module **200** may be configured to detect and decipher visual input signals, such as gestures, motions, sign language, or the like. Input module **200** may include each of a touch sensor, a visual detector, a microphone, and the like. In some embodiments, a plurality of input modules **200** are provided, with each input module **200** being directed to a different input mode.

[0046] Referring back to FIG. **1**, a schematic diagram of an external communication system **190** will be described according to an exemplary embodiment of the present subject matter. In general, external communication system **190** is configured for permitting interaction, data transfer, and other communications with appliance **100**. For example, this communication may be used to provide and receive operating parameters, cycle settings, performance characteristics, user preferences, user notifications, or any other suitable information for improved performance of appliance **100**. Additionally or alternatively, external communication system **190** may facilitate communication between appliance **100** and one or more additional appliances.

[0047] External communication system **190** permits controller **160** of appliance **100** to communicate with external devices either directly or through a network **192**. For example, a consumer may use a consumer device **194** to communicate directly with appliance **100**. For example, consumer devices **194** may be in direct or indirect communication with appliance **100**, e.g., directly through a local area network (LAN), Wi-Fi, Bluetooth, Zigbee, etc. or indirectly through network **192**. In general, consumer device **194** may be any suitable device for providing and/or receiving communications or commands from a user. In this regard, consumer device **194** may include, for example, a personal phone, a tablet, a laptop computer, or another mobile device.

[0048] In addition, a remote server **196** may be in communication with appliance **100** and/or consumer device **194** through network **192**. In this regard, for example, remote server **196** may be a cloud-based server **196**, and is thus located at a distant location, such as in a separate state, country, etc. In general, communication between the remote server **196** and the client devices may be carried via a network interface using any type of wireless connection, using a variety of communication protocols (e.g., TCP/IP, HTTP, SMTP, FTP), encodings or formats (e.g., HTML, XML), and/or protection schemes (e.g., VPN, secure HTTP, SSL).

[0049] In general, network **192** can be any type of communication network. For example, network **192** can include one or more of a wireless network, a wired network, a personal area network, a local area network, a wide area network, the internet, a cellular network, etc. According to an exemplary embodiment, consumer device **194** may communicate with a remote server **196** over network **192**, such as the internet, to provide user inputs, transfer operating parameters or performance characteristics, receive user notifications or instructions, etc. In addition, consumer device **194** and remote server **196** may communicate with appliance **100** to communicate similar information.

[0050] External communication system **190** is described herein according to an exemplary embodiment of the present subject matter. However, it should be appreciated that the exemplary functions and configurations of external communication system **190** provided herein are used only as examples to facilitate description of aspects of the present subject matter. System configurations may vary, other communication devices may be used to communicate directly or indirectly with one or more appliances, other communication protocols and steps may be implemented, etc. These variations and modifications are contemplated as within the scope of the present subject matter.

[0051] Now that the construction of a dishwasher appliance **100** and the configuration of controller **160** according to exemplary embodiments have been presented, an exemplary method **300** of operating a dishwasher appliance will be described. Although the discussion below refers to the exemplary method **300** of operating dishwasher appliance **100**, one skilled in the art will appreciate that the exemplary method **300** may be applicable to the operation of a variety of other appliances, for instance, appliances including doors (e.g., washing machine appliances, refrigerator appliances, etc.). In exemplary embodiments, the various method steps as disclosed herein may be performed by controller **160** or a separate, dedicated controller.

[0052] At step **302**, method **300** may include determining that the door is moved from the unlatched position to the latched position. As mentioned above, the door (e.g., door **116**) of the appliance (e.g., appliance **100**) may be movable between the latched (or closed) position and the unlatched (or open, ajar) position. According to some embodiments, a sensor (e.g., sensor **119**) may be incorporated. The sensor may be configured to determine a position of the door. The sensor may also be configured to measure an elapsed time of the door in the latched position or the unlatched position.

[0053] At step **304**, method **300** may include receiving, via the input module, a first input pattern of one or more input patterns. For instance, method **300** may activate the input module (e.g., input module **200**) after determining that the door is moved to the latched position. Accordingly, the input module may be activated to receive one of the plurality of inputs. As mentioned, the plurality of inputs may include one or more of touch inputs such as touch patterns, voice inputs such as voice commands, gestures such as swiping, signaling, or the like, or any combination of inputs.

[0054] As mentioned, the sensor may keep track of a length of time for which the door remains in the latched position after being moved from the unlatched position to the latched position. Method **300** may thus include determining that the door has been in the latched position for greater than a predetermined threshold length of time after being moved into the latched position. The predetermined threshold length of time may vary according to embodiments. According to some instances, the predetermined threshold length of time may be between about 4 seconds and about 10 seconds.

[0055] Upon determining that the door has been in the latched position for longer than the predetermined threshold length of time, method **300** may include deactivating the input module. For instance, method **300** may determine that an extended period of time has passed such that a washing operation is not desired (e.g., by the user). In order to avoid errant or unintentional inputs to the input module, the input module may be deactivated. In some instances, method **300** ignores any inputs to the input module after the predetermined threshold length of time has elapsed.

[0056] In the event that the door has been in the latched position for less than the predetermined threshold length of time, method **300** may receive the first input pattern. As mentioned, according to some examples, the first input pattern is a first touch input pattern. The touch input pattern may include a series of touch patterns, such as motions, swipes, length presses (e.g., maintaining contact with the input module for an extended period of time), or the like. Additionally or alternatively, the first input pattern may include a voice pattern, such as a voice command. Further still, the first input pattern may include one or more gestures, such as a wave, a hand motion or gesture, or the like. The first input pattern may be a unique input pattern.

[0057] The first input pattern may correspond to a particular washing cycle. In detail, the washing cycle or operation may be customized by a user pertaining to a set of cycle settings. For one example, the washing cycle is a full washing cycle requiring a full detergent supply, a maximum length of time, etc. Accordingly, the first input pattern may signal to the appliance (e.g., via the controller) that the full washing cycle is desired. As would be understood, multiple input patterns may be created and stored (e.g., within the appliance). Each input pattern may be a unique pattern. For instance, additional input patterns may correspond to different cycle settings pertaining to distinct washing operations (e.g., light wash, rinse only, low temperature, etc.).

[0058] The first input pattern may be preprogrammed. For instance, the first input pattern may be designed or set by a user. As such, the first input pattern may be customized by the user. The first input pattern may be programmed into the appliance. In some instances, a mobile application or app may be utilized to design, program, and store the first input pattern (and any subsequent input patterns). Each unique input pattern may be named, numbered, or otherwise identified within the memory of the appliance (or a memory of a connected device).

[0059] At step **306**, method **300** may include retrieving the first set of cycle settings corresponding to the first input pattern in response to receiving the first input pattern. As mentioned, the first input pattern may trigger a signal to incorporate a preset or predefined set of cycle settings or operational parameters for a requested washing operation. Thus, when the first input pattern is received (or recognized), the first set of cycle settings corresponding to the predetermined wash cycle (e.g., the full wash operation) may be retrieved, e.g., from a memory, and incorporated. According to some embodiments, the appliance may then enter a stand-by mode.

[0060] Method **300** may include receiving, via the input module, a second input pattern of the one or more input patterns after retrieving the first set of cycle settings. For instance, the second input pattern may be associated with initiating or starting the washing cycle. As mentioned, the second input pattern may be different from the first input pattern. According to some embodiments, the second input pattern includes a long touch input. For instance, as described above, the input module may be configured to receive a plurality of different input methods. The long touch input may correspond to a user maintaining physical contact with the input module for a predetermined length of time. In some instances, the predetermined length of time is between about 2 seconds and about 5 seconds. However, it should be understood that the predetermined length of time may vary according to specific embodiments, and the disclosure is not limited to the examples provided herein.

[0061] At step **308**, method **300** may include initiating a washing cycle after retrieving the first set of cycle settings. In detail, with the selected settings implemented, the washing cycle may begin.

The washing cycle may begin after receiving the second input pattern. However, in some instances, the first input pattern may include an automatic start feature. Accordingly, after receiving the first input pattern, the first cycle settings may be incorporated and the washing cycle may be initiated.
[0062] According to the disclosure, one or more shortcuts may be incorporated into an appliance to quickly and easily select desired washing cycles including customized settings. The shortcuts may be programmed into the appliance. The shortcuts may include one or more input patterns, such as touch patterns, voice patterns, gestures, or the like. The input patterns may be received by an input module. The input module may be provided apart from a traditional user interface. The input module may include a touch input positioned on a handle of the appliance. Accordingly, then the door is in a closed and latched position, a user may provide an input pattern to the input module to select and initiate a particular desired cycle.
[0063] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A dishwasher appliance comprising: a tub defining a receiving chamber; a door movably coupled to the tub to provide selective access to the receiving chamber, the door being movable between a latched position and an unlatched position; an input module provided on the door, the input module configured to receive one or more input patterns; and a controller operably coupled with the input module, the controller configured to perform an operation, the operation comprising: determining that the door is moved from the unlatched position to the latched position; receiving, via the input module, a first input pattern of the one or more input patterns after determining that the door is moved from the unlatched position to the latched position; retrieving a first set of cycle settings corresponding to the first input pattern in response to receiving the first input pattern; and initiating a washing cycle after retrieving the first set of cycle settings.
2. The dishwasher appliance of claim 1, further comprising: a sensor configured to sense a closed position of the door, wherein determining that the door is in the latched position comprises sensing, via the sensor, a position of the door.
3. The dishwasher appliance of claim 1, wherein the input module comprises a touch module configured to receive touch input patterns.
4. The dishwasher appliance of claim 1, wherein the operation further comprises: receiving, via the input module, a second input pattern of the one or more input patterns after retrieving the first set of cycle settings, wherein initiating the washing cycle is performed in response to receiving the second input pattern.
5. The dishwasher appliance of claim 4, wherein the second input pattern comprises a long touch input.
6. The dishwasher appliance of claim 1, wherein the input module comprises a microphone configured to receive voice input patterns.
7. The dishwasher appliance of claim 1, wherein the first input pattern corresponds to a predetermined wash cycle, the predetermined wash cycle comprising the first set of cycle settings.
8. The dishwasher appliance of claim 1, wherein the operation further comprises: activating the input module upon determining that the door is moved from the unlatched position to the latched position; determining that the door has been in the latched position for greater than a predetermined

threshold length of time after being moved to the latched position; and deactivating the input module upon determining that the door has been in the latched position for greater than the predetermined threshold length of time.

9. The dishwasher appliance of claim 1, wherein the input module is provided on a handle of the door.

10. A method of operating a dishwasher appliance, the dishwasher appliance comprising a door and an input module, the method comprising: determining that the door is moved from an unlatched position to a latched position; receiving, via the input module, a first input pattern of among a plurality of input patterns after determining that the door is moved from the unlatched position to the latched position; retrieving a first set of cycle settings corresponding to the first input pattern in response to receiving the first input pattern; and initiating a washing cycle after retrieving the first set of cycle settings.

11. The method of claim 10, wherein the dishwasher appliance further comprises: a sensor configured to sense a closed position of the door, wherein determining that the door is in the latched position comprises sensing, via the sensor, a position of the door.

12. The method of claim 10, wherein the input module comprises a touch module configured to receive touch input patterns.

13. The method of claim 10, further comprising: receiving, via the input module, a second input pattern among the plurality of input patterns after retrieving the first set of cycle settings, wherein initiating the washing cycle is performed in response to receiving the second input pattern.

14. The method of claim 13, wherein the second input pattern comprises a long touch input.

15. The method of claim 10, wherein the input module comprises a microphone configured to receive voice input patterns.

16. The method of claim 10, wherein the first input pattern corresponds to a predetermined wash cycle, the predetermined wash cycle comprising the first set of cycle settings.

17. The method of claim 10, further comprising: activating the input module upon determining that the door is moved from the unlatched position to the latched position; determining that the door has been in the latched position for greater than a predetermined threshold length of time after being moved to the latched position; and deactivating the input module upon determining that the door has been in the latched position for greater than the predetermined threshold length of time.

18. The method of claim 10, wherein the input module is provided on a handle of the door.